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Week 3: Image Segmentation

Ungraded Plugin: 3.1

Overview of Image Segmentation

6 min

Ungraded Plugin: 3.2

Segmentation by Humans

7 min

Practice Assignment: 3.2

Segmentation by Humans Self-Check Quiz

Submitted

Ungraded Plugin: 3.3

Segmentation as Clustering

5 min

Practice Assignment: 3.3

Segmentation as Clustering Self-Check Quiz

Submitted

Ungraded Plugin: 3.4

k-Means Segmentation

9 min

Practice Assignment: 3.4

k-Means Segmentation Self-Check Quiz

Submitted

Ungraded Plugin: 3.5

Mean-Shift Segmentation

10 min

Practice Assignment: 3.5

Mean-Shift Segmentation Self-Check Quiz

3h

Ungraded Plugin: 3.6

Graph-Based Segmentation

10 min

Practice Assignment: 3.6

Graph-Based Segmentation Self-Check Quiz

3h

Week 3 Quiz

Next Item →

3.4 k-Means Segmentation Self-Check Quiz

Your grade: 100%

100%

1 / 2 points

1. Which of the following options is correct for initializing the centroids of the k-Means clustering algorithm?

1 / 1 point

Select k random feature points as centroids

Select k uniformly distributed means in the range of the distribution

Perform k-means clustering on a subset of pixels and use them as initial means

All of the above

None of the above

Correct

Assignment details

Refer to 4:10 of the lecture

Submitted

Aug 15, 12:59 AM IST

Attempts

Unlimited

Try again

Help me practice

Let's chat

2. How does k-Means clustering get affected by a few noise pixels (distant outliers) in the image?

1 / 1 point

They always shift the centroid of the cluster towards the outliers

They always cause the centroid values of all the clusters (including those that do not have any outlier associated with them) to shift in their direction

Both A and B

None of the above

Correct

Like

Dislike

Report an issue

Your grade

100%

100%

View submission

See feedback

The outlier can only affect the centroid value of the cluster to which it is assigned. In some cases, if the outlier is too far away, the centroid can shift too far from the other pixels. Additionally, there are instances where the shifted centroid can influence other clusters, but it is not always guaranteed that all other clusters will be affected by the outliers.

https://www.coursera.org/learn/perception/assignment-submission/EC2bf3-4-k-means-segmentation-self-check-quiz/view-feedback

1/1